

TSU, Japan

WMO: 476510

Lat: 34.7342N

Lon: 136.5197E

Elev: 18

StdP: 101.10

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	0.9	-9.2	1.7	4.7	-7.6	2.0	4.9	12.4	6.4	11.5	5.9	4.0	280	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	33.8	25.2	32.5	25.3	31.4	25.0	26.7	31.3	26.2	30.6	25.7	29.9	4.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	29.5	25.0	20.1	28.9	24.5	19.5	28.5	84.0	31.3	81.5	30.7	79.5	30.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	10.0	8.8	-1.7	36.6	1.1	1.1	-2.5	37.4	-3.1	38.0	-3.7	38.6	-4.5	39.4		
			-3.7	27.8	1.1	0.6	-4.5	28.2	-5.2	28.5	-5.8	28.9	-6.6	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.7	6.1	9.1	14.3	19.2	22.8	27.2	28.3	24.7	19.1	13.2	8.2
	DBStd	8.24	2.12	2.41	2.75	3.02	2.33	2.30	2.41	1.96	2.57	2.65	2.79	2.57
	HDD10.0	375	134	113	52	4	0	0	0	0	0	4	68	
	HDD18.3	1654	392	343	287	124	17	0	0	0	22	154	315	
	CDD10.0	2767	1	3	23	133	285	385	533	568	441	283	101	12
	CDD18.3	1006	0	0	0	3	44	136	274	310	191	47	1	0
	CDH23.3	8724	0	0	1	13	138	665	2803	3582	1403	120	0	0
	CDH26.7	2952	0	0	0	1	18	132	1022	1442	329	8	0	0
Wind	WSAvg	3.8	4.3	4.2	4.2	4.0	4.0	3.6	3.5	3.7	3.7	3.4	3.4	3.9
Precipitation	PrecAvg	1694	54	67	111	135	174	208	176	151	292	171	101	54
	PrecMax	2298	154	151	220	237	433	318	341	431	579	531	284	120
	PrecMin	1010	2	23	41	43	67	84	48	19	49	29	10	3
	PrecStd	299	41	33	44	61	85	63	88	99	136	98	69	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	15.3	19.2	25.1	29.0	32.2	35.9	36.1	33.1	27.7	21.9	16.7
		MCWB	8.1	10.1	12.1	15.1	17.6	23.6	24.8	25.0	24.4	20.6	15.4	12.0
	2%	DB	11.9	12.9	16.8	22.2	26.3	29.8	33.8	34.1	31.0	26.0	20.2	15.1
		MCWB	7.3	8.4	11.1	14.7	17.8	22.2	25.2	25.3	23.9	19.7	15.2	10.3
	5%	DB	10.6	11.6	15.2	20.6	24.7	28.0	32.4	32.8	29.7	24.6	18.9	13.8
		MCWB	6.5	7.5	10.5	14.7	17.5	21.8	25.3	25.6	23.6	19.2	14.4	9.6
	10%	DB	9.4	10.4	13.9	19.1	23.3	26.7	31.2	31.8	28.5	23.4	17.9	12.6
		MCWB	5.5	6.6	9.6	14.1	17.3	21.3	25.0	25.3	23.2	18.4	13.6	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	11.9	14.7	19.6	21.8	25.0	27.3	27.7	26.3	23.2	17.9	14.0
		MCDB	11.9	13.5	16.5	21.5	24.5	29.7	32.1	32.2	29.8	25.6	19.4	15.2
	2%	WB	8.2	9.8	13.0	17.3	20.5	24.1	26.6	27.0	25.5	21.7	16.7	11.5
		MCDB	10.7	11.9	15.2	19.9	23.3	28.0	31.4	31.6	28.8	24.4	18.7	13.9
	5%	WB	6.9	8.4	11.7	16.1	19.5	23.3	26.0	26.5	25.0	20.4	15.6	10.2
		MCDB	9.9	10.8	14.1	18.7	22.6	26.7	30.7	30.9	28.1	23.3	17.9	13.0
	10%	WB	5.9	7.0	10.3	15.1	18.7	22.5	25.5	26.0	24.4	19.2	14.5	9.0
		MCDB	9.0	9.9	13.2	17.9	21.7	25.6	29.8	30.2	27.5	22.4	17.3	12.1
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	7.5	7.7	7.0	5.8	5.7	6.0	5.9	6.5	6.9	6.6
		MCDBR	7.8	8.2	9.3	9.6	9.5	8.0	7.8	7.4	7.1	7.4	7.5	7.9
		MCWBR	5.1	5.5	6.0	5.4	4.9	3.6	3.3	3.2	3.4	3.9	4.7	5.2
	5% WB	MCDBR	7.2	6.9	7.6	7.0	6.8	6.1	6.5	6.2	5.6	5.8	5.9	6.7
		MCWBR	5.4	5.7	6.4	5.2	4.7	3.2	3.2	3.2	3.1	4.0	4.8	5.4
Clear-Sky Solar Irradiance	taub	0.399	0.440	0.501	0.543	0.550	0.571	0.567	0.535	0.498	0.464	0.420	0.395	
	taud	2.163	2.063	1.918	1.867	1.917	1.938	1.997	2.072	2.146	2.158	2.188	2.187	
	Ebn at Noon	777	786	769	760	759	742	741	758	766	755	748	753	
	Edh at Noon	118	145	182	200	193	189	177	161	143	130	114	109	
All-Sky Solar Radiation	RadAvg	2.44	3.11	4.05	4.80	5.16	4.61	4.92	5.06	3.98	3.20	2.58	2.21	
	RadStd	0.16	0.24	0.27	0.48	0.57	0.44	0.66	0.44	0.37	0.34	0.24	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page